FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 15

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Recap *of previous lecture*

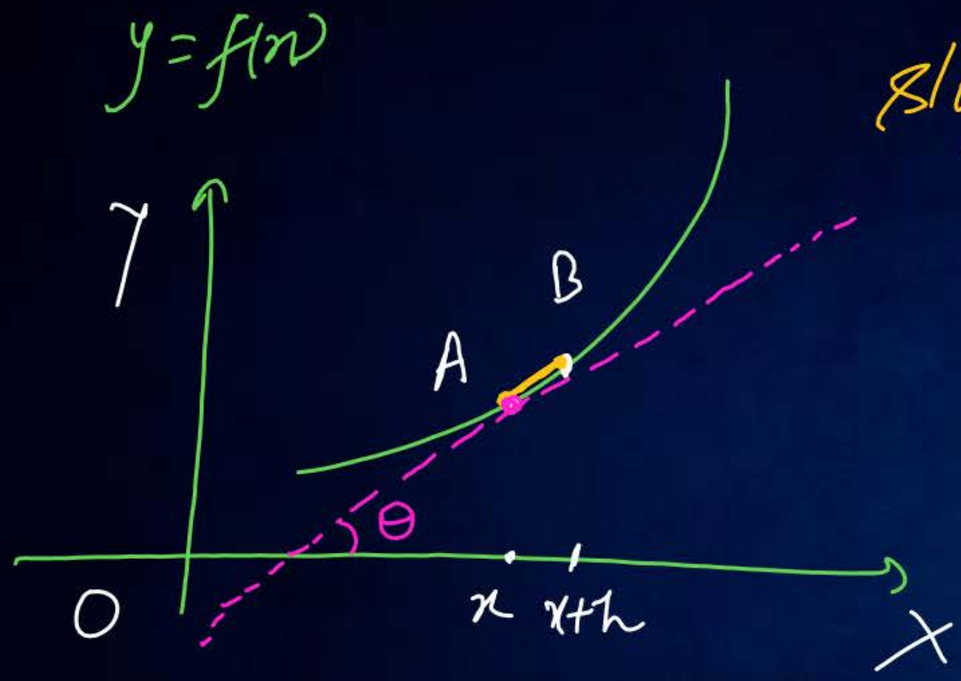
LIMITS



Topics *to be covered*

"DIFFERENTIATION"





Slope of straight line $AB = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(x+h) - f(x)}{(x+h) - (x)}$

$$= \frac{f(x+h) - f(x)}{h}$$

if we take B is very very close to A then we can take $h \to 0$, In that case AB converts into the tangent at point A of $f(x)$

$A(x, f(x))$,

$B(x+h, f(x+h))$

$\therefore$ Slope of tangent $= \tan\theta$

i.e Slope of AB = Slope of tangent $= \lim_{h \to 0} \left[\frac{f(x+h) - f(x)}{h} \right] = f'(x)$

Def^n: for $y = f(x)$

$$f'(x) = \boxed{\lim_{h \to 0} \left[\frac{f(x+h) - f(x)}{h}\right]} = \frac{dy}{dx}$$

$$f'(x) = \lim_{h \to 0} \left[\frac{f(x-h) - f(x)}{-h}\right] = \frac{dy}{dx}$$

Above is called Derivative by using 1st principle Method.

eg $y = e^x = f(x)$ then $f'(x) = \lim_{h \to 0} \left[\frac{e^{x+h} - e^x}{h}\right] = e^x \lim_{h \to 0} \left(\frac{e^h - 1}{h}\right) = e^x(1) = e^x$

eg $y = f(x) = \tan^{-1} x$ then $f'(x) = \lim_{h \to 0} \left[\frac{\tan^{-1}(x+h) - \tan^{-1} x}{h}\right] = \lim_{h \to 0} \left[\frac{\tan^{-1}\left[\frac{x+h-x}{1+(x+h)x}\right]}{h}\right]$

$\left(\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)\right)$

$\lim_{x \to 0} \left(\frac{\tan x}{x}\right) = 1$

$$f'(x) = \lim_{h\to 0}\left[\frac{\tan^{-1}\left(\frac{h}{1+x^2+hx}\right)}{h}\right] = \lim_{h\to 0}\frac{\tan^{-1}\left(\frac{h}{1+x^2+hx}\right)}{\left(\frac{h}{1+x^2+hx}\right)} \times \frac{1}{(1+x^2+hx)}$$

$$= \lim_{h\to 0}\left[\frac{\tan^{-1}\left(\frac{h}{1+x^2+hx}\right)}{\left(\frac{h}{1+x^2+hx}\right)}\right] \cdot \lim_{h\to 0}\left(\frac{1}{1+x^2+hx}\right)$$

$$\left[\because \lim_{x\to 0}\left(\frac{\tan^{-1}x}{x}\right) = 1\right]$$

$$\frac{d}{dx}\left(\tan^{-1}x\right) = 1 \times \frac{1}{1+x^2+0} = \frac{1}{1+x^2}$$

(M-II) $y = \tan^{-1}x \Rightarrow \tan y = x \Rightarrow \frac{d}{dx}(\tan y) = \frac{d}{dx}(x) \Rightarrow \sec^2 y \cdot \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y}$

$$\text{So } \frac{dy}{dx} = \frac{1}{1+x^2}$$

Some Standard Results :-

(1) $\boxed{\frac{d}{dx}(a^x) = a^x \log_e a}$

eg $\frac{d}{dx}(e^x) = e^x$

(2) Power Formula: $\boxed{\frac{d}{dx}(x^a) = a x^{a-1}}$

$\frac{d}{dx}(k) = 0$, $\frac{d}{dx}(x) = 1$, $\frac{d}{dx}(x^2) = 2x$

$\frac{d}{dx}(x^3) = 3x^2$, $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$, $\frac{d}{dx}|x| = \frac{x}{|x|} = \frac{|x|}{x}$

(3) $\boxed{\frac{d}{dx}(\log_a x) = \frac{1}{x \log_e a}}$ eg $\frac{d}{dx}(\log_e x) = \frac{1}{x}$

(4) $\frac{d}{dx}(\sin x) = \cos x$, (6) $\frac{d}{dx}(\tan x) = \sec^2 x$

(5) $\frac{d}{dx}(\cos x) = -\sin x$, (7) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

(8) $\frac{d}{dx}(\sec x) = \sec x \tan x$

(9) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

(10) $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$, $-1 < x < 1$

(11) $\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$, $-1 < x < 1$

(12) $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$, $-\infty < x < \infty$

(13) $\frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2}$, $-\infty < x < \infty$

(14) $\frac{d}{dx}(\sec^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}}$, $|x| > 1$

(15) $\frac{d}{dx}(\operatorname{cosec}^{-1}x) = \frac{-1}{|x|\sqrt{x^2-1}}$, $|x| > 1$

(16) $\frac{d}{dx}|x| = \frac{x}{|x|} = \frac{|x|}{x}$, $x \neq 0$

(17) Product formula →

(i) $\frac{d}{dx}(fg) = fg' + gf'$

(ii) $\frac{d}{dx}(fgh) = fgh' + fg'h + f'gh$

(18) Quotient formula →

$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{gf' - fg'}{(g)^2}$

(19) $\frac{d}{dx}(\text{const}) = 0$ (20) $\frac{d}{dx}[c \cdot f(x)] = c \cdot \frac{d}{dx}(f(x))$

(21) $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$

$$\boxed{\frac{d}{dx}(x^a) = ax^{a-1}}$$

eg $\frac{d}{dx}(x^3) = 3x^2$

eg $\frac{d}{dx}(x^4) = 4x^3$

eg $\frac{d}{dx}(x) = 1 \cdot x^{1-1} = 1$

eg $\frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}\left(x^{\frac{1}{2}}\right) = \frac{1}{2}x^{\frac{1}{2}-1}$

$= \frac{1}{2}x^{-1/2} = \boxed{\frac{1}{2\sqrt{x}}}$

eg $\frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right) = ? =$

eg $\frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right) = \frac{d}{dx}\left(x^{-\frac{1}{2}}\right) = -\frac{1}{2}x^{-\frac{1}{2}-1} = \frac{-1}{2}x^{-3/2}$

eg $\frac{d}{dx}(|x|) = ? = \frac{d}{dx}\left(\sqrt{x^2}\right) = \frac{1}{2\sqrt{x^2}}\frac{d}{dx}(x^2) = \frac{x}{\sqrt{x^2}} = \frac{x}{|x|}$

$= \frac{x|x|}{|x|^2} = \frac{x|x|}{x^2} = \frac{|x|}{x}$; $x \neq 0$

$\boxed{|x| = \sqrt{x^2} = \begin{cases} -x, & x < 0 \\ +x, & x \geq 0 \end{cases}}$

eg $\frac{d}{dx}\left(\frac{1}{x}\right) = ? = \frac{d}{dx}(x^{-1}) = -1 \cdot x^{-1-1} = -1 \cdot x^{-2} = \boxed{\frac{-1}{x^2}}$

eg $\frac{d}{dx}(k) = ? = k\frac{d}{dx}(x^0) = k \times 0 \cdot x^{0-1} = 0$

(12) Chain Rule: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \frac{d}{dx} g(x) = f'\{g(x)\} \cdot g'(x)$

$\frac{d}{dx}(e^{x^3}) = ? = e^{x^3} \cdot \frac{d}{dx}(x^3) = e^{x^3}(3x^2)$

Q. If $y = e^x \log \sqrt{x} \tan x$ then $\frac{dy}{dx} = ?$

$= e^x \log x^{\frac{1}{2}} \cdot \tan x$

$y = \frac{1}{2} \underset{f}{e^x} \cdot \underset{g}{\log x} \cdot \underset{h}{\tan x}$

$\frac{dy}{dx} = \frac{1}{2}\left\{ e^x \log x \sec^2 x + e^x \cdot \frac{1}{x} \tan x + e^x \log x \cdot \tan x \right\}$

Q. $y = (1+x)(1+x^2)(1+x^4)(1+x^8)$ then $\frac{dy}{dx} = ?$

$y = \frac{(1-x)(1+x)(1+x^2)(1+x^4)(1+x^8)}{(1-x)}$

$= \frac{(1-x^2)(1+x^2)(1+x^4)(1+x^8)}{1-x}$

$= \frac{(1-x^4)(1+x^4)(1+x^8)}{1-x} = \frac{(1-x^8)(1+x^8)}{1-x}$

$y = \frac{1-x^{16}}{1-x}$

If $y = \frac{1-x^{16}}{1-x}$ then

$$\frac{dy}{dx} = \frac{(1-x)\cdot\frac{d}{dx}(1-x^{16}) - (1-x^{16})\frac{d}{dx}(1-x)}{(1-x)^2}$$

$$= \frac{(1-x)(0-16x^{15}) - (1-x^{16})(0-1)}{(1-x)^2}$$

$$= \frac{-16x^{15} + 16x^{16} + 1 - x^{16}}{(1-x)^2}$$

$$= \frac{-16x^{15} + 1 + 15x^{16}}{(1-x)^2}$$

Q: $y = \log(\sin x^2) \Rightarrow \frac{dy}{dx} = ?$

Sol: $\frac{dy}{dx} = \frac{1}{\sin x^2}\cdot\frac{d}{dx}(\sin x^2)$

$$= \frac{1}{\sin x^2}\cdot(\cos x^2)\cdot\frac{d}{dx}(x^2) = \cot x^2(2x)$$

Q: $y = \log(\sec x + \tan x)$ then $y' = ?$

$$\frac{dy}{dx} = \frac{1}{\sec x + \tan x}\frac{d}{dx}(\sec x + \tan x)$$

$$= \frac{1}{\sec x + \tan x}(\sec x\tan x + \sec^2 x) = \sec x$$

(*) $\frac{d}{dx}(\underset{I}{\sin x}\cdot\underset{II}{e^x}) = ? = \sin x\,(e^x) + e^x\,(\cos x)$ //

$\frac{d}{dx}(xy) = x\frac{d}{dx}(y) + y\frac{d}{dx}(x) = x\frac{dy}{dx} + y(1)$ //

$\frac{d}{dy}(xy) = ? = x\frac{d}{dy}(y) + y\frac{d}{dy}(x) = x(1) + y\frac{dx}{dy}$

(*) $\boxed{\frac{dy}{dx}\cdot\frac{dx}{dy} = 1} \Rightarrow \frac{dy}{dx} = \frac{1}{(dx/dy)}$ Learn,

Q. $\sin y = x \sin(a+y)$ then $\frac{dy}{dx} = ?$

$\Rightarrow x = \frac{\sin y}{\sin(a+y)}$

Diff both sides w.r.to y

$$\frac{dx}{dy} = \frac{d}{dy}\left[\frac{\sin y}{\sin(a+y)}\right]$$

$$= \frac{\sin(a+y)\cdot\frac{d}{dy}(\sin y) - \sin y \frac{d}{dy}\sin(a+y)}{(\sin(a+y))^2}$$

$$= \frac{\sin(a+y)\cos y - \sin y \cos(a+y)}{\sin^2(a+y)}$$

$$\frac{dx}{dy} = \frac{\sin(a+y-y)}{\sin^2(a+y)} = \frac{\sin a}{\sin^2(a+y)}$$

Hence $\frac{dy}{dx} = \frac{1}{(dx/dy)} = \frac{\sin^2(a+y)}{\sin a}$

Log Diff i-e

eg if $y = x^x$ then $\frac{dy}{dx} = ?$

Sol: $\log_e y = \log_e(x^x) = x(\log_e x)$

$$\frac{d}{dx}(\log_e y) = \frac{d}{dx}[x \cdot \log x]$$

(I) (II)

$$\frac{d}{dy}(\log y) \cdot \frac{dy}{dx} = x\frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = x\left(\frac{1}{x}\right) + \log x$$

$$\frac{dy}{dx} = y \cdot (1 + \log x) = \boxed{x^x(1+\log x)}$$

Q: if $y = x^x + x^a + a^x + a^a$ then $y' = ?$

$$\frac{dy}{dx} = x^x(1+\log x) + ax^{a-1} + a^x \log_e a + 0$$

Q: $y = \log_{10} x + \log_x 10 + \log_x x + \log_{10} 10$

$$y = \frac{\log_e x}{\log_e 10} + \frac{\log_e 10}{\log_e x} + 1 + 1$$

$$\frac{dy}{dx} = \frac{1}{\log_e 10}\left(\frac{1}{x}\right) + \log_e 10\left(\frac{-1}{(\log x)^2} \cdot \frac{1}{x}\right) + 0 + 0$$

Q: if $x^y = y^x$ then $\frac{dy}{dx} = ?$

Sol: $\log_e(x^y) = \log_e(y^x)$

$y \log x = x \log y$

$\frac{d}{dx}(\underset{I\ II}{y \log x}) = \frac{d}{dx}(\underset{I\ II}{x \log y})$

$y \frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x) = x \frac{d}{dx}(\log y) + \log y \frac{d}{dx}(x)$

$y\left(\frac{1}{x}\right) + \log x (1) = x\left(\frac{d}{dy}(\log y) \cdot \frac{dy}{dx}\right)^{*} + \log y (1)$

$\frac{y}{x} + \log x = x\left(\frac{1}{y} \cdot \frac{dy}{dx}\right) + \log y$

$\frac{y}{x} + \log x - \log y = \frac{x}{y}\frac{dy}{dx}$

$-\frac{x}{y}\frac{dy}{dx} = -\frac{y}{x} + \log y - \log x$

$-\frac{x}{y}\frac{dy}{dx} = -\frac{y}{x} + \log\left(\frac{y}{x}\right)$

$\frac{dy}{dx} = 1 - \frac{y}{x}\log\left(\frac{y}{x}\right)$ //

Explicit fun^n $y = f(x)$ & Implicit fun^n: $f(x,y) = c$

Diff of Implicit fun^n →

Eg $x^2 + 2xy + y^3 = 42$ then $\frac{dy}{dx} = ?$

$f(x,y) = c$

Sol: Diff both sides w.r.to. 'x'

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(2xy) + \frac{d}{dx}(y^3) = \frac{d}{dx}(42)$$

$$2x + 2\frac{d}{dx}(xy) + \frac{d}{dy}(y^3)\cdot\frac{dy}{dx} = 0$$

$$2x + 2\left[x\frac{dy}{dx} + y(1)\right] + (3y^2)\frac{dy}{dx} = 0$$

$$(2x + 3y^2)\frac{dy}{dx} = -y - 2x$$

$$\frac{dy}{dx} = -\left(\frac{y+2x}{2x+3y^2}\right)$$ //

$x \neq 0$

Q. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ then $\frac{dy}{dx} = ?$

M-I

$$\frac{d}{dx}\left[x\sqrt{1+y}\right] + \frac{d}{dx}\left(y\sqrt{1+x}\right) = 0$$

. — — — — — . Lengthy

M-II

$$x\sqrt{1+y} = -y\sqrt{1+x}$$

$$x^2(1+y) = y^2(1+x)$$

$$x^2 + x^2y - y^2 - y^2x = 0$$

$$(x^2 - y^2) + (x^2y - y^2x) = 0$$

$$(x-y)(x+y) + xy(x-y) = 0$$

$$(x-y)[x+y+xy] = 0$$

is either $x - y = 0$ or $x + y + xy = 0$

is $y = x$ or $y = \frac{-x}{1+x}$ Explicit

Not possible

So $\frac{dy}{dx} = \frac{d}{dx}\left(\frac{-x}{1+x}\right) = -\left[\frac{(1+x)(1) - x(1)}{(1+x)^2}\right]$

$$y' = \frac{-1}{(1+x)^2}$$

⊛ Diff of Infinite Series :-

Q: $y=\sqrt{\sin x+\sqrt{\sin x+\sqrt{\sin x+\cdots\infty}}}$ then $\frac{dy}{dx}=?$

Sol: $y=\sqrt{\sin x+y} \Rightarrow y^2=\sin x+y$

ie $y^2-y=\sin x$ ——— (1)

$$\frac{d}{dx}(y^2-y)=\frac{d}{dx}(\sin x)$$

$$\frac{d}{dy}(y^2-y)\frac{dy}{dx}=\cos x$$

$$(2y-1)\frac{dy}{dx}=\cos x \Rightarrow \frac{dy}{dx}=\frac{\cos x}{2y-1}$$

Q: $y=x^{x^{x^{x^{\cdots\infty}}}}$ then $\frac{dy}{dx}=?$

Sol: $y=x^y \Rightarrow \log_e y=\log_e(x^y)$

$$\log y=y\log x \quad \text{——— (1)}$$

$$\frac{d}{dx}(\log y)=\frac{d}{dx}(y\log x)$$

$$\frac{1}{y}\cdot\frac{dy}{dx}=y\left(\frac{1}{x}\right)+\log x\frac{d}{dx}(y)$$

$$\left(\frac{1}{y}-\log x\right)\frac{dy}{dx}=\frac{y}{x} \Rightarrow y'=\frac{y^2}{x}\left(\frac{1}{1-y\log x}\right)$$ //

Diff of parametric func^n:- if $x=f(\theta)$ & $y=g(\theta)$ then $\frac{dy}{dx}=?$ $=\boxed{\frac{dy}{d\theta}\cdot\frac{d\theta}{dx}}$

$\Downarrow$ $\Downarrow$

$\frac{dx}{d\theta}=?$ $\quad$ $\frac{dy}{d\theta}=?$

Q: if $x=a(\theta-\sin\theta)$ & $y=a(1-\cos\theta)$ then $\frac{dy}{dx}=?$

Sol: $\frac{dx}{d\theta}=a[1-\cos\theta]$ & $\frac{dy}{d\theta}=a(0+\sin\theta)$

Hence $\frac{dy}{dx}=\frac{dy}{d\theta}\cdot\frac{d\theta}{dx}=(a\sin\theta)\left(\frac{1}{a(1-\cos\theta)}\right)$

$=\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}=\cot\frac{\theta}{2}$

Q: if $x=a\cos^3t$, $y=a\sin^3t$ then $\frac{dy}{dx}=?$

$\frac{dx}{dt}=a\left(3\cos^2t(-\sin t)\right)=-3a\sin t\cos^2t$

$\frac{dy}{dt}=a\left(3\sin^2t\cdot\cos t\right)=3a\sin^2t\cos t$

$\frac{dy}{dx}=\frac{dy}{dt}\cdot\frac{dt}{dx}=\frac{3a\sin^2t\cos t}{-3a\sin t\cos^2t}=-\tan t$

(*) Diff of func^n w.r.to another func^n

Q: Diff $u=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.to $v=\tan^{-1}x$; $-1<x<1$

Let $\tan\theta = x \Rightarrow \theta = \tan^{-1}x$

$\Rightarrow u = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) = \sin^{-1}(\sin 2\theta) = 2\theta$

& Let $v = \tan^{-1}x = \theta$

$\because u = 2\theta$ & $v = \theta$

$\frac{du}{d\theta} = 2$ & $\frac{dv}{d\theta} = 1$

$\frac{du}{dv} = \frac{du}{d\theta}\cdot\frac{d\theta}{dv}$

$= (2)\left(\frac{1}{1}\right) = 2$

Q: Diff $u=x^x$ w.r.to $v=(x\log x)$

Sol: $u = x^x \Rightarrow \frac{du}{dx} = x^x(1+\log x)$

& let $v = x\log x \Rightarrow \frac{dv}{dx} = x\left(\frac{1}{x}\right)+\log x\,(1)$

$= 1+\log x$

$\frac{du}{dv} = \frac{du}{dx}\cdot\frac{dx}{dv}$

$= \left[x^x(1+\log x)\right]\left(\frac{1}{1+\log x}\right) = x^x$

(Pg) if $x^2+y^2=t-\frac{1}{t}$ & $x^4+y^4=t^2+\frac{1}{t^2}$ then $\frac{dy}{dx}=?$

—— (1) ——— (2)

Sol: $(x^2+y^2)^2=\left(t-\frac{1}{t}\right)^2$

$x^4+y^4+2x^2y^2=t^2+\frac{1}{t^2}-2(t)\left(\frac{1}{t}\right)$

$x^4+y^4+2x^2y^2=x^4+y^4-2$

$2x^2y^2=-2 \Rightarrow y^2=-\frac{1}{x^2}$

$\frac{d}{dx}(y^2)=-\frac{d}{dx}\left(\frac{1}{x^2}\right)$

$2y\frac{dy}{dx}=-\{-2x^{-2-1}\}$

$\frac{dy}{dx}=\frac{1}{y\cdot x^3}$


Thank
YOU